Multi-Property Inverse Design of Crystalline Materials via Energy-Based Matching and Extrapolation

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Inverse design of crystalline materials requires identifying structures that simultaneously optimize multiple properties such as thermodynamic stability, electronic band gap, and mechanical stiffness. Because these targets conflict, the designer's object of interest is not a single optimum but a trade-off surface, which demands models capable of learning joint representations across multiple correlated properties and of remaining reliable at the edges of their training distributions where novel materials are typically found. To address this, we propose training directly in a joint multi-property space rather than predicting targets in isolation. We introduce MEX-Multi, a multi-property framework that equips each target with an independent energy-based model (EBM) head and utilizes a derivative-free optimizer (DFO) for inference. All properties share a single SE(3)-equivariant backbone (EquiformerV2), which is the only parameter set the properties share: each head receives the same structural embedding and is coupled to the others solely through the gradients that flow back into that backbone. We evaluate MEX-Multi on a 5,000-structure subset of the Materials Project MP20 dataset across three simultaneous targets: formation energy, band gap, and Voigt–Reuss–Hill bulk modulus (K_{VRH}). The framework achieves mean absolute errors (MAE) of 0.110 eV/atom, 0.381 eV, and 0.087 $\log_{10}$GPa, with Spearman correlations of 0.957, 0.716, and 0.969. A controlled comparison against single-property baselines—sharing identical backbones, heads, losses, and data splits—shows that joint multi-property training improves *every* metric for *every* property, reducing MAE by 68%, 42%, and 59% respectively. This isolates transfer through the shared representation, rather than added capacity, as the primary source of the performance gain. Beyond prediction, applying NSGA-II non-dominated sorting over per-property DFO candidate sets exposes Pareto-optimal trade-off surfaces that we compare directly against the true DFT-measured front. Finally, embedding-space retrieval against the held-out Alexandria database confirms this shared representation transfers effectively for formation energy and band gap, while explicitly quantifying limitations in transferring bulk modulus characteristics.

CCS Concepts: • **Computing methodologies** → **Neural networks**; *Learning latent representations*; • **Applied computing** → **Physical sciences and engineering**.

Additional Key Words and Phrases: inverse design, materials informatics, energy-based model, multi-property prediction, multi-objective optimization, equivariant neural networks, extrapolation, Materials Project

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1 Introduction

Functional materials are rarely evaluated on a single physical metric. For instance, a viable photovoltaic absorber must simultaneously exhibit an electronic band gap between 1.1 and 1.7 eV, thermodynamic stability near the convex hull, and sufficient mechanical stiffness to withstand manufacturing. Similarly, a thermal barrier coating requires a precise balance of stiffness, thermal conductivity, and oxidation resistance. While optimizing for any one of these requirements yields a vast number of candidate structures, satisfying them concurrently is severely constraining. Furthermore, these target properties often inherently conflict: the same atomic structural motifs that increase mechanical stiffness frequently disrupt the desired band gap. Consequently, a materials designer does not typically seek a single universal optimum. Instead, the goal is to discover a *trade-off surface*—the complete set of jointly attainable, mutually non-dominated property vectors—along with the specific crystal structures that realize them.

Density functional theory (DFT) calculations can evaluate many of these material properties reliably, but at a cost that makes exhaustive search over even a modest candidate library infeasible: a single structural relaxation and total-energy calculation already costs on the order of 10^2–10^3 CPU-hours per compound [17], a full elastic tensor adds roughly another twenty strained-cell relaxations on top of that [8], and even after applying charge-neutrality and electronegativity filters the enumerable space of chemically plausible quaternary inorganic compositions alone already exceeds 10^{12} candidates [7] — dwarfing the ∼184,000 structures characterised in the ICSD and the tens of thousands computed to date in the Materials Project [18]. As a result, practical exploration is confined to narrowly constrained chemical spaces chosen in advance, which is precisely the wrong strategy when the goal is to find materials that are unusual on several axes at once.

As surrogates for DFT calculations, machine learning models have relaxed this bottleneck for individual properties [4, 6, 23, 27, 35], but they usually predict the properties one at a time. Composing K independently trained single-property models into a multi-property screen is unsatisfying for two reasons. First, it discards the physical correlations among targets — the interplay between band gap and formation energy in oxides [5, 30] is information that each model is prevented from using. Second, the resulting error structure is opaque: K separately calibrated models make errors that are uncorrelated with one another and with the underlying physics, so the composite screen offers no principled account of which regions of the joint property space it can be trusted to describe.

Multi-task materials models address the first objection, but only partially. MT-CGCNN [30] and multi-fidelity graph networks [5] attach several heads to a shared representation and demonstrate real gains on correlated targets. In these architectures, however, properties interact only *through the loss*: the shared trunk accumulates gradients from every task, but the representation of a structure never depends on the observed value of any property. Nothing in the forward pass tells the network that this particular crystal is stiff, or metallic, while it is forming its embedding.

A second and more fundamental difficulty is *extrapolation*. Screening datasets are, by construction, poor in exactly the structures a designer wants: those with extreme or simultaneously optimal values across several targets. A direct regressor $f_\theta : \mathcal{G} \to \mathbb{R}$ trained by a pointwise loss has no mechanism for emitting a value outside the range it was trained to output, and its behaviour — and the gradients derived from it — degrades fastest at the frontier where design happens. The Matching-based EXtrapolation (MEX) framework [28] responds to this in the single-property

setting by reframing prediction as *energy-based matching*: rather than regressing y from $\mathbf{x}$, MEX learns a joint energy $E_\theta(\mathbf{x}, y)$ under which the correct property value is the argmax and all alternatives are ranked below it, and recovers predictions at test time with a derivative-free optimizer (DFO) that refines candidate values against the learned energy landscape. Because the model *scores* values rather than emitting them, it can assign high energy to a value it never saw as a training label.

This work extends MEX to the multi-property setting. MEX-Multi equips each target with an independent energy head over a single shared equivariant backbone, and — this is the central design choice — conditions that backbone on the observed values of the *other* targets during training, so that inter-property structure enters the representation itself rather than only the gradient signal. Crucially, MEX-Multi performs design in *property space* rather than structure space: it does not generate new crystals, but uses the trained energy heads to propose attainable property vectors for structures in an existing library, and non-dominated sorting [9] to expose the trade-off surface among them. This is cheaper than generative inverse design and, more importantly, directly auditable — because every proposal remains attached to a real structure with known DFT properties, the proposed front can be compared against the true measured front, and the learned embedding can be validated by retrieval against a held-out database [12].

Our contributions are the following claims, each stated with the evidence that supports it.

(1) **Training one shared equivariant backbone against K energy-based heads outperforms training the same architecture one property at a time.** Holding backbone, energy-head architecture, losses, data split and seeds fixed and varying only whether the model is trained on one property or three, joint training improves every metric for every property, reducing MAE by 68%, 42% and 59% for formation energy, band gap and bulk modulus (Table 3). Because capacity and training budget are constant, the gain is attributable to correlation the shared representation can exploit, not to added parameters.

(2) **The negative sampling distribution must be matched to each property's range individually.** Locally sampled NCE negatives shape a sharp energy peak near the ground truth but leave the landscape undefined elsewhere, which is sufficient for narrow-range targets and fails for wide-range ones, causing the DFO to converge on a spurious peak. Uniformly drawn InfoNCE negatives resolve this without altering the treatment of narrower targets (Section 6.3).

(3) **The same energy heads support design, not just prediction.** Repeated DFO passes followed by non-dominated sorting yield Pareto-optimal trade-off surfaces that we compare directly against the true DFT-measured front (Figure 3).

(4) **Predictive accuracy and embedding geometry are not interchangeable evidence.** Embedding-space retrieval against the held-out Alexandria database recovers property distributions closer to the test set than random retrieval for formation energy and band gap — and for band gap this requires joint training, since no single-property embedding beats chance. For bulk modulus, retrieval fails to beat random despite strong predictive accuracy on that same property, a limitation we report rather than omit (Section 6.5).

We evaluate on a 5,000-structure subset of the Materials Project MP20 dataset [18], drawn from the 9,803 MP20 structures that carry all three property labels simultaneously (Section 5), across formation energy, band gap and Voigt–Reuss–Hill bulk modulus.

2 Related Work

MEX-Multi sits at the intersection of three literatures that have developed largely independently: property prediction, from which we take our backbone; multi-task learning, whose coupling mechanism we depart from; and inverse design, whose goal we share but whose generative approach we do not. We review each in turn, then the energy-based formulation that connects them.

2.1 Machine Learning for Materials Property Prediction

Learned surrogates for density functional theory (DFT) now span a wide range of architectures. The crystal graph convolutional neural network (CGCNN) [35] established the dominant representation for periodic solids—atoms as nodes, neighbours within a cutoff as edges—and demonstrated that a single graph model could predict formation energy, band gap and elastic moduli to useful accuracy. MEGNet [4] generalised this to a unified framework covering both molecules and crystals with global state variables, and ALIGNN [6] added explicit bond-angle information through a line graph, improving accuracy on properties sensitive to local geometry.

A parallel line of work has established that encoding Euclidean symmetry directly in the architecture, rather than learning it from data, substantially improves data efficiency. SchNet [31] introduced continuous-filter convolutions over interatomic distances; DimeNet [13] added directional message passing; and Equiformer [21] and its successor EquiformerV2 [22] combined equivariant attention with high-degree spherical harmonic representations, the latter scaling to $l_{\max} = 6$ at tractable cost. The same inductive bias underpins the current generation of universal interatomic potentials, including NequIP [2], MACE [1] and M3GNet [3]. Standardised benchmarks such as Matbench [11] have made cross-architecture comparison routine.

What these models share is the framing: each learns a direct mapping $f_\theta : \mathcal{G} \to \mathbb{R}$ trained by minimising a pointwise regression loss. As Section 1 argues, this framing is precisely what degrades under extrapolation, since a smooth global regressor has no mechanism for placing a prediction outside the range of values it was trained to output. We retain EquiformerV2 as our backbone, but replace the regression head with the energy-based matching formulation of MEX [28]. Retaining that backbone raises the question of how the three targets should share it, which is the subject of the next subsection.

2.2 Multi-Property and Multi-Task Learning for Materials

The observation that materials properties are correlated, and that this correlation is exploitable, is not new. MT-CGCNN [30] attaches several prediction heads to a shared CGCNN representation and reports up to 8% error reduction on correlated targets—formation energy, band gap and Fermi energy—with the largest gains in the low-data regime. Multi-fidelity graph networks [5] make a related argument from a different direction, using abundant low-fidelity (PBE) band gaps as an auxiliary signal to sharpen predictions of scarce high-fidelity targets, and reducing experimental band gap error by 22–45%.

MEX-Multi shares the central assumption of this line of work: properties interact through the shared trunk, which receives gradient contributions from every task, while the representation of a structure never depends on the observed value of any property. What differs is the setting rather than the coupling mechanism. The multi-task models above attach regression heads to the shared trunk, so each task is still a direct structure-to-value mapping and inherits the extrapolation behaviour that motivates this work; MEX-Multi instead attaches K energy-based matching heads, so multi-task transfer and matching-based extrapolation are combined in one model. That combination raises a question

with no analogue in multi-task regression—how negatives should be sampled when the targets differ by an order of magnitude in range—which Section 6.3 addresses. Table 3 isolates the benefit of the shared trunk itself by holding backbone, heads, loss and split fixed and varying only whether the model is trained on one property or three.

A second gap is evaluative. Multi-task materials models are almost always assessed on in-distribution accuracy; their behaviour under extrapolation, and whether the shared representation is useful for anything beyond prediction, is rarely tested. We report both, via the extrapolation-motivated MEX objective and the embedding-space retrieval study of Section 6.

2.3 Inverse Design Methods

The multi-task models above improve prediction accuracy over a shared representation, but they stop short of design: none proposes candidates, and none reasons about the jointly attainable region. We turn now to methods that do, and to the respect in which MEX-Multi's approach to design differs from theirs.

Computational inverse design divides broadly into generative and search-based families. Generative methods learn a distribution over structures and sample from it under property constraints. FTCP [29] introduced an invertible crystallographic representation permitting decoding from a learned latent space; CDVAE [34] combined a variational autoencoder with a diffusion decoder that respects periodicity; DiffCSP [19] jointly diffuses the lattice and fractional coordinates under periodic $E(3)$ equivariance; and MatterGen [36] scales this paradigm to a pretrained diffusion model that is fine-tuned through adapter modules to steer generation toward chemistry, symmetry, and mechanical, electronic or magnetic constraints. Search-based methods instead explore a fixed or combinatorially generated candidate space, using evolutionary algorithms [26] or, more recently, large-scale surrogate-driven screening [24].

Two limitations are relevant here. First, conditioning on *several* properties at once remains comparatively underexplored: most conditional generators are fine-tuned per constraint, and satisfying multiple constraints jointly is typically handled by filtering samples after generation rather than by the model itself. Second, generative approaches produce structures that must subsequently be validated by DFT, which is expensive and re-introduces the bottleneck the surrogate was meant to remove.

MEX-Multi occupies a different position. It does not generate structures; it performs design in *property space* over an existing candidate library, using the trained energy heads to propose attainable property vectors for a given structure and NSGA-II [9] non-dominated sorting to expose the trade-off surface among them. This is cheaper than generation and directly auditable: because candidates are attached to real structures, the proposals can be compared against the true DFT-measured front (Figure 3) and the embedding can be validated by retrieval against a held-out database [12]. Both the extrapolation behaviour and the auditability just described follow from the energy-based formulation rather than from the multi-property extension, so we review that formulation's origins before developing it.

2.4 Energy-Based Models in Scientific Applications

Energy-based models [20] specify an unnormalised density $p(\mathbf{x};\theta) \propto e^{E_\theta(\mathbf{x})}$, trading tractable likelihoods for architectural freedom. The resulting intractable partition function has motivated a family of training strategies—Markov chain Monte Carlo with Langevin dynamics [10], noise-contrastive estimation [16], and contrastive objectives with uniformly drawn negatives such as InfoNCE [33]—surveyed comparatively in [32]. In the physical sciences, EBMs have been used to sample equilibrium states of many-body systems [25], where the energy formulation aligns naturally with the underlying physics.

Most directly relevant is the use of EBMs for *regression*. Gustafsson *et al.* [14] model the conditional density $p(y|\mathbf{x};\theta)$ with a network that scores $(\mathbf{x}, y)$ pairs, and predict by maximising that score over y at test time; a follow-up study [15] compares training objectives and finds NCE with locally sampled negatives to be the strongest. MEX [28] brings this formulation to materials, replacing gradient-based refinement with a derivative-free optimiser and demonstrating that the matching framing extrapolates better than direct regression.

Our contribution to this line is the multi-property setting, which raises an issue absent when a single target is modelled: the negative sampling distribution must be matched to the range of each property individually. Local Gaussian negatives shape a sharp energy peak near the ground truth but leave the landscape undefined elsewhere, which is adequate for narrow-range targets and fails for wide-range ones—in our case K_{VRH}, whose normalised range spans roughly 22 units. We therefore train that property with uniformly drawn InfoNCE negatives (Eq. (9)) while retaining local NCE for E_{form} and E_g, a per-property choice that has no analogue in single-target EBM regression.

The target is the Voigt–Reuss–Hill *bulk* modulus, K_{VRH} — the Materials Project elasticity endpoint we query returns K_{VRH}, not the shear modulus K_{VRH}, and we correct the symbol accordingly throughout this paper. The range quoted in Section 5, $[-1.01, +10.58]$ in $\log_{10}$GPa, is not a mislabelling of normalised units: it is exactly the raw range of the 9,803-structure pool described there. It is, however, set entirely by a handful of known artefacts in the Materials Project elasticity data — 47 of the 9,803 structures exceed 1,000 GPa, and the single largest, at 3.8×10^{10} GPa, is many orders of magnitude beyond any physically realised bulk modulus (diamond, among the stiffest known solids, is ≈443 GPa), while the median (83.5 GPa) is physically unremarkable. We report the data as extracted rather than silently re-filtering it, but flag the extreme tail explicitly: it is what stretches both the ∼22 normalised units quoted above and the DFO search bounds of Eq. (10), which are anchored to the training set's raw extremes.

3 Background: MEX for Single-Property Prediction

Having positioned MEX-Multi against the prediction, multi-task and inverse-design literatures, we now state the single-property formulation it extends. This section is a compressed restatement of MEX [28]; Section 4 introduces what is new.

Let $\mathcal{G}$ denote a crystal structure (a periodic graph of atoms) and let $y \in \mathbb{R}$ be a scalar material property (e.g., formation energy). A standard regressor learns $f_\theta : \mathcal{G} \to \mathbb{R}$. MEX instead learns a joint energy function

$$E_\theta(\mathbf{x}, y) \in \mathbb{R}, \tag{1}$$

where $\mathbf{x} = \phi(\mathcal{G}) \in \mathbb{R}^D$ is the crystal embedding produced by an equivariant graph neural network ϕ, and the predicted property is

$$\hat{y} = \arg\max_{y'} E_\theta(\mathbf{x}, y'). \tag{2}$$

3.0.1 *Energy architecture.*

The energy function decomposes as

$$E_\theta(\mathbf{x}, y) = \langle f_x(\mathbf{x}),\ f_{xy}(\mathbf{x}, y) \rangle, \tag{3}$$

where f_x is a linear projection of the crystal embedding and f_{xy} combines the crystal embedding with a learned encoding of y.

3.0.2 NCE training objective. To train the energy function, MEX uses a Noise-Contrastive Estimation (NCE) loss. Negative samples $\tilde{y}_1, \ldots, \tilde{y}_K$ are drawn from a local Gaussian mixture around the ground-truth value y^*:

$$\tilde{y}_i = y^* + \epsilon_i, \quad \epsilon_i \sim \mathrm{GMM}(0, \boldsymbol{\sigma}), \tag{4}$$

and the NCE loss is

$$\mathcal{L}_{\mathrm{NCE}} = -\mathbb{E}\left[\log \frac{e^{E_\theta(\mathbf{x}, y^*) - \log q(y^*)}}{e^{E_\theta(\mathbf{x}, y^*) - \log q(y^*)} + \sum_{i=1}^{K} e^{E_\theta(\mathbf{x}, \tilde{y}_i) - \log q(\tilde{y}_i)}}\right]. \tag{5}$$

An auxiliary Euclidean loss enforces geometric alignment between the structural and property embeddings:

$$\mathcal{L}_{\mathrm{euc}} = \left\| \hat{f}_x(\mathbf{x}) - \hat{f}_{xy}(\mathbf{x}, y^*) \right\|^2, \tag{6}$$

where $\hat{\cdot}$ denotes ℓ_2 normalisation.

3.0.3 Derivative-Free Optimizer (DFO). At inference, maximising the energy over y from Eq. (2) is performed by a sample-and-refine procedure:

(1) Draw S candidates $\{y^{(s)}\}_{s=1}^{S}$ uniformly from $[y_{\mathrm{lb}}, y_{\mathrm{ub}}]$.
(2) For T iterations: compute $E_\theta(\mathbf{x}, y^{(s)})$ for all candidates, resample proportionally to softmax(E), then perturb with Gaussian noise of decreasing scale $\sigma_t = \sigma_0 \cdot \rho^t$.
(3) Return the candidate with the highest energy.

4 Method: Multi-Property MEX (MEX-Multi)

Single-property MEX supplies the energy-matching machinery and the derivative-free inference procedure, but it fits one model per target, so nothing constrains the representations of two correlated properties to agree — each target is learned in isolation, exactly as in the direct regressors it replaces. This section supplies that coupling by training a single shared backbone against K energy heads simultaneously, and derives the two consequences that follow from it: a per-property choice of negative sampling distribution, and a design procedure over the resulting K heads.

Figure 2 gives a complete overview of MEX-Multi, from the shared backbone through both the training path (left, blue) and the two evaluation paths used throughout Section 6: point-estimate prediction and Pareto-optimal design (centre, orange), and embedding-space retrieval validation (right, green). The remainder of this section derives each labelled component in detail.

4.1 Multi-Property Architecture

Let $\{y_k\}_{k=1}^{K}$ be K target properties for a crystal $\mathcal{G}$. MEX-Multi uses a single shared backbone ϕ to produce one crystal embedding $\mathbf{x} = \phi(\mathcal{G}) \in \mathbb{R}^D$, and K independent EBM heads, one per property. Head k owns a label encoder $f_y^{(k)} : \mathbb{R} \to \mathbb{R}^D$ and a scoring network $g^{(k)} : \mathbb{R}^{2D} \to \mathbb{R}$, and assigns the energy

$$E_\theta^{(k)}(\mathbf{x}, y_k) = g^{(k)}\left(\left[\, \mathbf{x};\ f_y^{(k)}(y_k) \,\right]\right), \tag{7}$$

where $[\,\cdot\,;\,\cdot\,]$ denotes concatenation. Following the score module of single-property MEX (Section 3), $g^{(k)}$ is a three-layer residual MLP terminating in a scalar projection, and $f_y^{(k)}$ is a linear map followed by a SiLU nonlinearity. Both are instantiated separately for each property, so no parameters are shared between heads.

4.1.1 What the properties share. The only parameters common to all K targets are those of the backbone ϕ. The embedding $\mathbf{x}$ passed to head k is identical to the one passed to every other head and does not depend on the observed

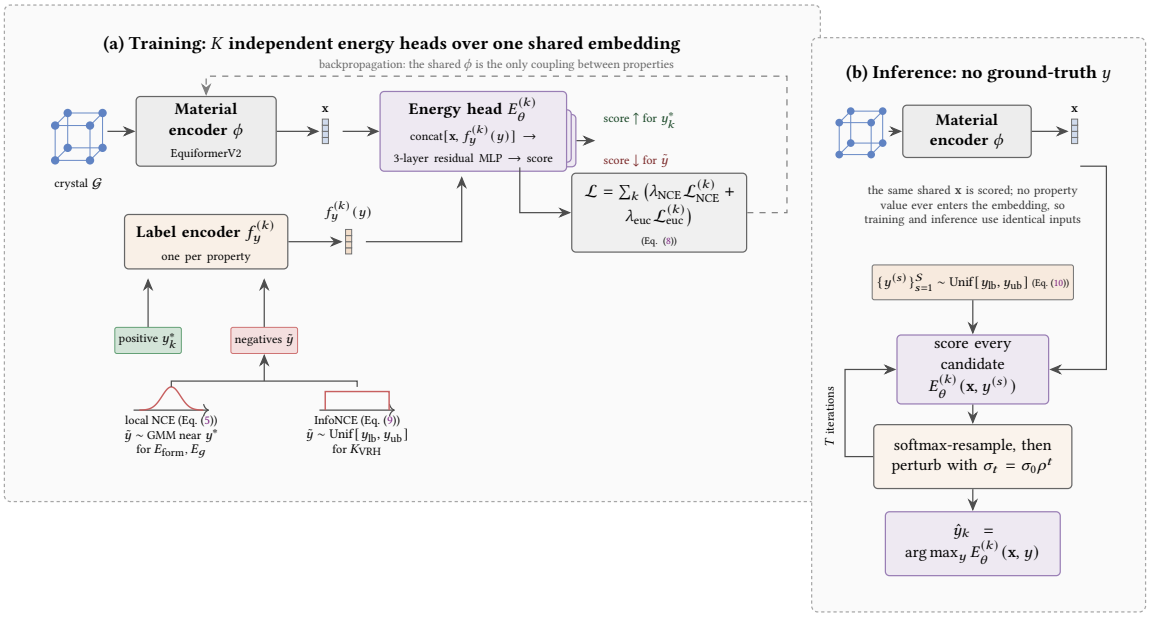


Fig. 1. The MEX-Multi framework. **(a) Training.** A crystal $\mathcal{G}$ is embedded once by the shared EquiformerV2 backbone ϕ, giving $\mathbf{x}$. Each property k owns a label encoder $f_y^{(k)}$ and an energy head: the candidate value is encoded, concatenated with $\mathbf{x}$, and scored by a three-layer residual MLP. Heads are trained contrastively to raise the score of the true value y_k^* relative to negatives — locally sampled Gaussian-mixture negatives for the narrow-range E_{form} and E_g (Eq. (5)), and uniformly drawn negatives spanning the full property range for the wide-range K_{VRH} (Eq. (9)). The per-property losses are summed (Eq. (8)) and backpropagated through ϕ, which is the only parameter set the properties share: no property value enters the embedding itself, so inter-property structure is transferred through the shared backbone rather than through the input to any head. **(b) Inference.** The same $\mathbf{x}$ is scored, so training and inference see identical inputs. The derivative-free optimiser draws S candidates within the bounds of Eq. (10), then repeatedly resamples them in proportion to their energy and perturbs them with a shrinking noise scale; after T iterations the highest-energy candidate is returned as $\hat{y}_k$.

value of any property; the value y_k enters only through head k's own label encoder. Inter-property structure therefore reaches the model in one way: the summed loss of Eq. (8) backpropagates through ϕ, so the representation is shaped simultaneously by all K objectives. Figure 1 makes this coupling explicit, and Table 3 measures what it is worth by holding the architecture fixed and varying only the number of properties trained on.

A consequence worth stating plainly, because it is what makes the evaluation in Section 6 clean, is that training and inference see exactly the same input to every head. No quantity available only at training time enters $\mathbf{x}$, so there is no train/test distribution shift to correct for and no possibility of label information leaking into a prediction — a risk that any scheme conditioning the shared representation on observed property values would have to address explicitly.

4.2 Loss Functions

The total training loss sums per-property contributions:

$$\mathcal{L} = \sum_{k=1}^{K} \left(\lambda_{\text{NCE}}^{(k)} \mathcal{L}_{\text{NCE}}^{(k)} + \lambda_{\text{euc}}^{(k)} \mathcal{L}_{\text{euc}}^{(k)} \right). \tag{8}$$

4.2.1 InfoNCE for wide-range properties. Local NCE (Eq. (5)) samples negatives within ±0.3 normalised units of y^*, which trains a sharp local energy peak but leaves the global energy landscape undefined. For wide-range properties (e.g., K_{VRH}, whose normalised range spans ∼23 units), the DFO can fail at inference by landing on a training-data peak far from the validation target.

To address this, we use InfoNCE for K_{VRH}, which draws negatives uniformly from the full normalised range $[y_{\text{lb}}, y_{\text{ub}}]$:

$$\mathcal{L}^{(k)}_{\text{InfoNCE}} = -\mathbb{E}\left[\log \frac{e^{E^{(k)}_\theta(\mathbf{x}, y^*)}}{\sum_{j=0}^{K} e^{E^{(k)}_\theta(\mathbf{x}, \tilde{y}_j)}}\right], \quad \tilde{y}_j \sim \text{Uniform}(y_{\text{lb}}, y_{\text{ub}}). \tag{9}$$

This forces the model to correctly rank y^* against *any* alternative in the property range, producing a globally consistent energy landscape.

4.3 DFO Bounds

A critical implementation detail is the choice of DFO search bounds $[y_{\text{lb}}, y_{\text{ub}}]$. We set bounds in normalised space as:

$$y_{\text{lb}} = y^{\text{train}}_{\text{min}} - \delta, \quad y_{\text{ub}} = y^{\text{train}}_{\text{max}} + \delta, \quad \delta = 2.0, \tag{10}$$

where $y^{\text{train}}_{\text{min/max}}$ are the normalised extremes of the training set. This ensures the search region closely tracks the data distribution regardless of normaliser variance, avoiding the pathological effect of a fixed raw-space floor (e.g., −10 eV) producing a 24-unit negative excursion for low-stdev properties like K_{VRH} ($\sigma = 0.49$).

4.4 Pareto-Optimal Inverse Design

With per-property optimisation and its search bounds now specified, the multi-property design problem reduces to a question of aggregation: how to combine K independent point estimates into the single trade-off surface that Section 1 identified as the designer's actual object of interest. The front is over *structures*, not over an optimiser's sampling noise: a structure has one formation energy, not a distribution of them, so the candidates being compared must be different structures' predictions, never repeated stochastic re-runs of one optimiser on a single structure. At inference, the model generates Pareto-optimal candidate structures via:

(1) For each structure m in the candidate library (of size M), run one DFO pass per property to obtain its point-estimate property vector $\hat{\mathbf{y}}_m \in \mathbb{R}^K$ — the same point estimate evaluated in Table 2.
(2) Assemble the cost matrix $C \in \mathbb{R}^{M \times K}$ from these M vectors, one row per structure. Lower is treated as better for all three properties as predicted (formation energy, band gap and K_{VRH} alike; we apply no per-property sign adjustment), so the front favours structures that are simultaneously most negative in formation energy and lowest in band gap and K_{VRH}.
(3) Apply NSGA-II non-dominated sorting [9] to C to extract the Pareto front: the subset of the M library structures not dominated by any other structure in the library.

This enables practitioners to browse the Pareto-optimal trade-off between conflicting constraints, evaluated in Section 6.4.

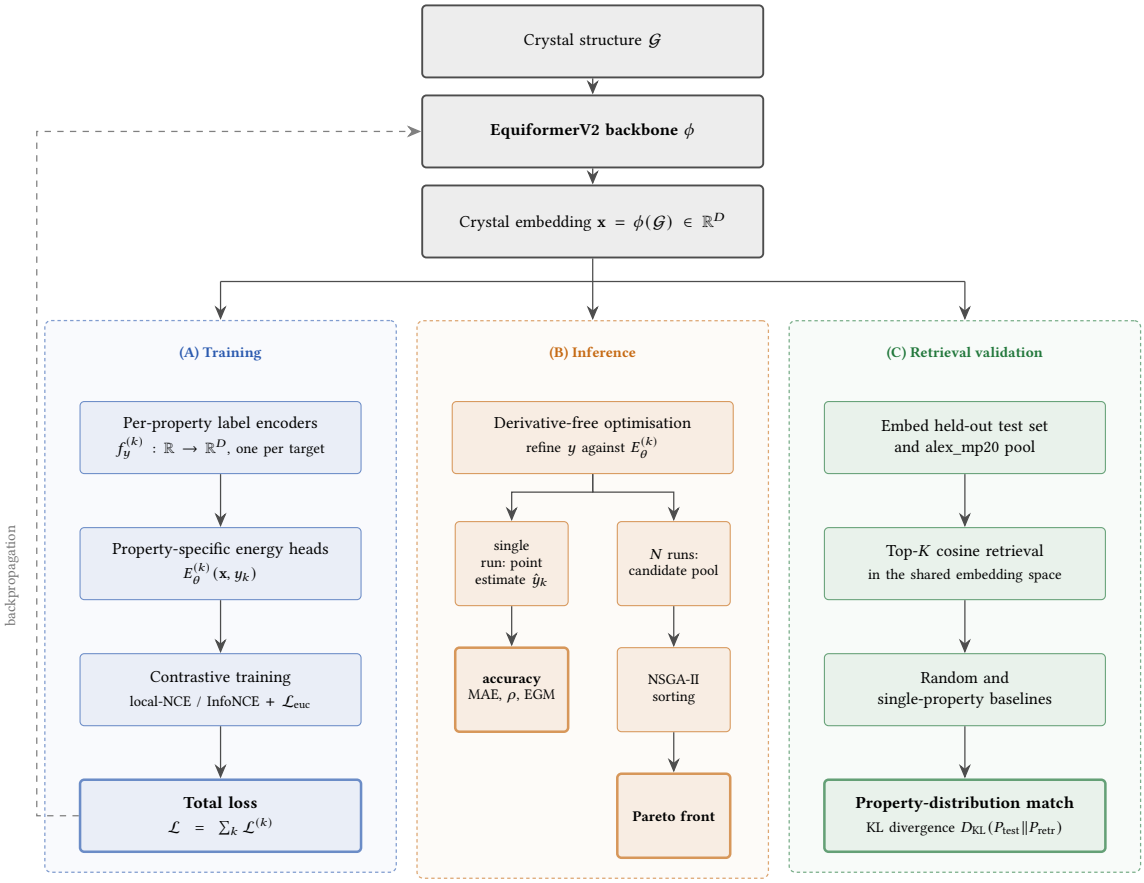


Fig. 2. MEX-Multi end-to-end workflow. A single EquiformerV2 backbone ϕ (8 layers, l_{max}=4, 128 spherical channels, 20-NN edges with a 12 Å cutoff, ≈28M parameters) maps each periodic crystal graph to one embedding $\mathbf{x}$ shared by all three stages. **(A) Training:** each property k has its own label encoder and energy head, which scores the shared $\mathbf{x}$ against a candidate value for that property (Eq. (7)); no property value enters $\mathbf{x}$ itself, so the heads are coupled only through the backbone they share (Section 4.1.1, Figure 1). Local-NCE negatives are used for E_{form} and E_g and InfoNCE negatives for K_{VRH} (Eq. (5), Eq. (9)), and the summed loss (Eq. (8), coefficients in Table 1) backpropagates through the shared backbone. **(B) Inference:** the heads score the same $\mathbf{x}$ used in training, so no train/test input shift arises; the same derivative-free optimiser – S=2000 samples refined over T=5 iterations against the trained heads – drives both read-outs: a single run per property gives the per-structure point estimate evaluated both in Table 2/Table 3 and, pooled as one vector per test-library structure, in NSGA-II non-dominated sorting across the library to yield the structure-level Pareto front (Figure 3). **(C) Retrieval validation:** the same backbone embeds the 250 held-out test structures and the ≈34,500-candidate alex_mp20 pool; top-K=10 cosine retrieval, pooled over 3 seeds, is compared against a random baseline and three single-property variants by the Kullback–Leibler divergence between the test and retrieved property distributions, with bootstrap confidence intervals (Figure 4). Full hyperparameters are given in Section 4.

5 Experimental Setup

Two features of the evaluation follow directly from the claims of Section 1: the split and seeds are held fixed across the single- and multi-property settings so that Table 3 varies only the number of properties trained on, and the retrieval pool is drawn from a database disjoint from all MP20 splits so that Section 6.5 tests transfer rather than memorisation.

5.1 Dataset

We use the Materials Project MP20 dataset [18], comprising approximately 45,000 inorganic crystalline structures. Three properties are predicted simultaneously:

- **Formation energy** (E_{form}, eV/atom): thermodynamic stability indicator; range $[-4.47, +0.08]$ eV/atom.
- **Band gap** (E_g, eV): electronic property; range $[0, 17.65]$ eV.
- **Voigt–Reuss–Hill bulk modulus** (K_{VRH}, $\log_{10}$ GPa): mechanical stiffness; range $[-1.01, +10.58]$, driven by a small number of data artefacts rather than physically realised stiffness (Section 2).

Each property is independently normalised to zero mean and unit variance using training-set statistics.

Elastic-tensor coverage in Materials Project is far sparser than formation energy or band gap: intersecting MP20 against the K_{VRH} records available from Materials Project's elasticity endpoint yields 9,803 structures carrying all three labels simultaneously, an order of magnitude below the full 45k pool. Every result in this paper — the point-estimate metrics of Table 2–Table 3, the Pareto-front evaluation of Section 6.4, and the retrieval evaluation of Section 6.5 — is computed on the same fixed 5,000-structure random subsample of this pool (seed 42), split 4,500/250/250 (90/5/5); the 250-structure test split is the held-out set throughout. This is not the MP20 dataset's own official train/val/test partition [18], but a fixed, reproducible split of the 9,803-structure three-label pool described above.

5.2 Backbone Model

We use EquiformerV2 [22] as the molecular encoder ϕ. The model is configured with 8 transformer layers, 128 spherical channels, spherical harmonics up to degree $l_{\text{max}} = 4$, and 20 nearest-neighbour edges within a 12 Å cutoff radius. The total backbone has approximately 28 million parameters.

5.3 Training Details

- **Optimiser:** AdamW, lr = 10^{-4}, no weight decay.
- **Scheduler:** ReduceLROnPlateau (factor 0.8, patience 10 epochs).
- **Batch size:** 64 structures.
- **EMA decay:** 0.999.
- **Gradient clipping:** $\|\nabla\|_2 \leq 10$.
- **Early stopping:** patience 30 epochs on $\overline{\text{MAE}}$.
- **Hardware:** Single NVIDIA V100 16 GB.

Per-property DFO settings: 500 training negatives, 2000 inference samples, 5 refinement iterations, initial noise scale 0.5 (decay factor 0.5 per iteration).

5.4 Loss Coefficients

Table 1. Loss coefficients per property

Property	Loss type	$\lambda_{\text{NCE/InfoNCE}}$	λ_{euc}
E_{form}	NCE	1.0	0.25
E_g	NCE	1.0	0.25
K_{VRH}	InfoNCE	1.0	0.25

5.5 Evaluation Metrics

We report mean absolute error (MAE), Spearman rank correlation ρ, and the error geometric mean (EGM). For n test structures with predictions $\hat{y}_i$ and ground truth y_i,

$$\mathrm{EGM} = \left(\prod_{i=1}^{n} \max\left(|\hat{y}_i - y_i|,\ \epsilon\right) \right)^{1/n} = \exp\left(\frac{1}{n} \sum_{i=1}^{n} \log \max\left(|\hat{y}_i - y_i|,\ \epsilon\right) \right), \tag{11}$$

with a floor $\epsilon = 10^{-10}$ applied only to guard against $\log 0$ if an error is ever exactly zero; across all three properties and all test structures we checked, no prediction error is ever exactly zero (model outputs are continuous), so this floor never actually activates in practice — it is a numerical safeguard, not the mechanism behind any of the reported numbers. MAE and EGM are both absolute-error measures and ρ is a rank-only measure, so the three need not agree on which property is best predicted — a dissociation that recurs for band gap throughout Section 6.

The real mechanism behind band gap's low EGM in Table 2 is the ground-truth distribution, not error flooring: 172 of the 250 test structures (68.8%) are metallic, with a true band gap of exactly 0 eV. This fraction is a fixed property of the test set, independent of model or seed. Because MEX-Multi's band-gap predictions regress toward this dominant near-zero mode (Section 6), most of these 172 structures also receive a very small (though essentially never exactly zero) predicted error. A geometric mean is dominated by its smallest values — log of a small number is a large negative contribution to the average that defines Eq. (11), whereas the same small value barely moves an arithmetic mean like MAE — so a large population of near-zero errors pulls EGM far below MAE specifically for band gap, in a way that does not happen for formation energy or K_{VRH}, whose ground truth is not concentrated at zero.

6 Results

6.1 Multi-Property Prediction Accuracy

Table 2 reports test-set MAE, Spearman rank correlation ρ, and error geometric mean (EGM) on the mp20_5k benchmark (mean ± std over 3 training seeds). A single-task literature-SOTA comparison (CGCNN [35], MEGNet [4], ALIGNN [6]) is omitted here: those published figures were obtained on the full ~45k-structure MP20 split, not the 5k-structure subset used throughout this paper, so they are not directly comparable and we leave a properly matched single-task baseline (retrained on the same 5k split) to future work.

Table 2. Test-set performance of MEX-Multi on the mp20_5k benchmark (mean ± std over 3 seeds). MEX-Multi predicts all three properties jointly with a single model.

Property	MAE↓	ρ↑	EGM↓
E_{form} (eV/atom)	0.110 ± 0.007	0.957 ± 0.000	0.055 ± 0.007
E_g (eV)	0.381 ± 0.039	0.716 ± 0.011	0.021 ± 0.004
K_{VRH} ($\log_{10}$GPa)	0.087 ± 0.004	0.969 ± 0.002	0.023 ± 0.002
Average MAE	0.193	—	—

The formation-energy Spearman's 0.957 ± 0.000 is not a rounding or reporting artefact: the three per-seed values are 0.95695, 0.95691 and 0.95713 (mean 0.95700, std 0.00012), genuinely agreeing to within about one part in a thousand. This is far tighter than band gap's cross-seed spread on the same metric (std 0.0114, roughly two orders of magnitude larger, Table 2), which is consistent with why: formation energy's training-set values are continuous and spread

across a wide range ([−4.47, +0.08] eV/atom, Section 5), so its rank ordering is well-determined and reproduces almost identically across training runs, while band gap's ground truth is dominated by exact ties at $E_g = 0$ (Section 5.5), and how a Spearman correlation breaks ties among a large tied block depends more sensitively on the specific (and seed-dependent) predicted values within that block.

MEX-Multi attains high rank-ordering fidelity for formation energy ($\rho = 0.957$) and K_{VRH} ($\rho = 0.969$), and a markedly lower ρ for band gap (0.716), reflecting the known difficulty of DFT band gap prediction: metallic structures ($E_g = 0$) dominate the distribution and regression towards the mean is expected even for well-trained models. Band gap's EGM (0.021) is nonetheless the lowest of the three properties despite its weaker Spearman — the error geometric mean is dominated by the many near-zero-gap (metallic) structures the model predicts accurately in absolute terms, even where it struggles to rank-order the smaller population of wide-gap structures correctly; MAE and EGM in Table 2 should therefore be read as absolute-error measures and Spearman as a complementary, rank-only measure, rather than expecting all three metrics to agree on which property is "best" predicted.

6.2 Effect of Joint Multi-Property Training: Single- vs. Multi-Property MEX

Table 2 establishes MEX-Multi's absolute accuracy; it does not by itself show that training the shared backbone against all three objectives at once (Section 4.1.1) is what produces that accuracy, as opposed to incidental differences in training budget or architecture capacity. To isolate the effect of joint training specifically, we compare MEX-Multi against *single-property* MEX models: the same backbone, energy-head architecture, loss functions, and training procedure, trained on one property at a time so the embedding is never exposed to the other two. Both settings use the identical mp20_5k train/val/test split and the same 3 seeds, so the only variable that changes between rows of Table 3 is whether the model sees one property or all three during training.

Table 3. Single-property MEX vs. MEX-Multi (multi-property), same backbone and mp20_5k split, mean ± std over 3 seeds.

Property	Model	MAE	Spearman	EGM
Form. Energy	Single	0.345 ± 0.019	0.753 ± 0.037	0.197 ± 0.012
	Multi	0.110 ± 0.007	0.957 ± 0.000	0.055 ± 0.007
Band Gap	Single	0.660 ± 0.021	0.650 ± 0.026	0.240 ± 0.034
	Multi	0.381 ± 0.039	0.716 ± 0.011	0.021 ± 0.004
Bulk Modulus	Single	0.211 ± 0.032	0.803 ± 0.085	0.108 ± 0.020
	Multi	0.087 ± 0.004	0.969 ± 0.002	0.023 ± 0.002

Joint multi-property training improves every metric for every property, with no exceptions. MAE drops by 68% for formation energy (0.345 → 0.110), 59% for bulk modulus (0.211 → 0.087), and 42% for band gap (0.660 → 0.381). Formation energy and bulk modulus also see the largest Spearman gains (+0.204 and +0.166 respectively), while band gap's Spearman improves only modestly (+0.066) despite band gap having the single largest relative EGM improvement of the three properties (91%, 0.240 → 0.021) — consistent with the MAE/EGM-vs-Spearman dissociation noted in Section 5.5 and Section 6 above, and reinforcing that joint training mainly helps the model get closer in absolute terms on the dominant near-zero-gap population rather than resolving the harder rank-ordering of wide-gap outliers.

This result is the controlled counterpart to the extrapolation argument motivating this work (Section 1): with only 5k training structures, a single-property model sees relatively few examples at the extremes of its one target's range, whereas MEX-Multi's backbone is shaped by gradients from all three objectives for every structure in the split, so each property's extremes are represented in a trunk that the other two also have to explain. Since the three properties are

not independent (e.g. structures with extreme bulk modulus are also structurally atypical in ways that affect formation energy), this shared supervision appears to partially compensate for the small per-property sample size — exactly the low-data, correlated-property regime the introduction identifies as the core motivation for extending MEX beyond single-property training.

6.3 Ablation: InfoNCE vs. NCE for K_{VRH}

Section 6.2 isolates the contribution of joint training; it leaves open whether the second design choice — per-property negative sampling — is similarly load-bearing or merely a convenience. Since range-matched negative sampling is claimed as a contribution in Section 1, it requires its own controlled test.

[**TODO:** run the controlled ablation: identical configuration, with K_{VRH} trained under local NCE vs. InfoNCE, all else fixed. `configs/mex-multi/mp20_5k_gvrh_nce_ablation.yml` (identical to `mp20_5k.yml` except y_{gvrh}'s loss is `nce` instead of `info-nce`), `run_scripts/train_mp20_5k_gvrh_nce_ablation_slurm.sh` (trains the same 3 seeds), and `predict_mp20_5k_gvrh_nce_ablation.sh` (predicts + computes MAE/ρ/EGM per seed) are ready to run — this still needs actual GPU training, not just inference, so the numbers are not yet available. Once run, report final test MAE/ρ/EGM across the 3 seeds against Table 2's K_{VRH} row, and pull validation MAE curves from Weights&Biases to show the DFO failure mode directly. Ideally also report the same swap for one narrow-range property, to show the choice is range-dependent rather than uniformly favouring InfoNCE]

6.4 Pareto Front Visualisation

The metrics above characterise MEX-Multi as a predictor. The design claim of Section 1 — that the same energy heads expose a trade-off surface — requires a different read-out of the same optimiser, which we evaluate here.

Figure 3 shows the corrected, structure-level Pareto front described in Section 4: one point-estimate property vector per test structure (the same point estimate evaluated in Table 2), non-domination computed across the 250-structure test library, overlaid with the *true* non-dominated front computed the same way but over the test structures' actual DFT-measured properties. Each row of Figure 3 is one of the three independently trained seeds: the model's own front contains 38, 39 and 42 structures for seeds 0, 1 and 2 respectively — 15–17% of the 250-structure library — while the true front, depending only on the real test-set structures and not on the model, is identical in every row at n=18. Unlike the within-structure resampling front this replaces, every point plotted here is a real, distinct test structure: the blue and orange markers are directly comparable populations, both drawn from the same 250 structures, so the front sizes above are directly interpretable as "what fraction of the test library the model judges simultaneously best," not an artefact of how many samples were drawn.

6.4.1 Comparison against the true test-set front. The 18-point true Pareto front reveals a pattern that holds identically across all three rows of Figure 3, and it is more specific than "the model misses rare outliers": the model's own front is heavily concentrated at near-zero band gap in every seed, while the true front has genuine spread across band gap, including a material at $E_g \approx 6.8$ eV and the outlier at $E_g \approx 17.7$ eV (essentially the dataset's documented maximum, 17.65 eV, Section 5). No structure in any of the three seeds' model fronts reaches a band gap comparable to either point. This is consistent with, and gives a more precise explanation for, the band-gap-specific weakness already noted in Section 6: MEX-Multi's band-gap predictions regress toward the near-zero, metallic mode of the training distribution (Spearman ρ=0.716, the weakest of the three properties), so almost no test structure's *predicted* band gap is high enough for that structure to survive non-domination on the wide-gap end of the trade-off surface — even though

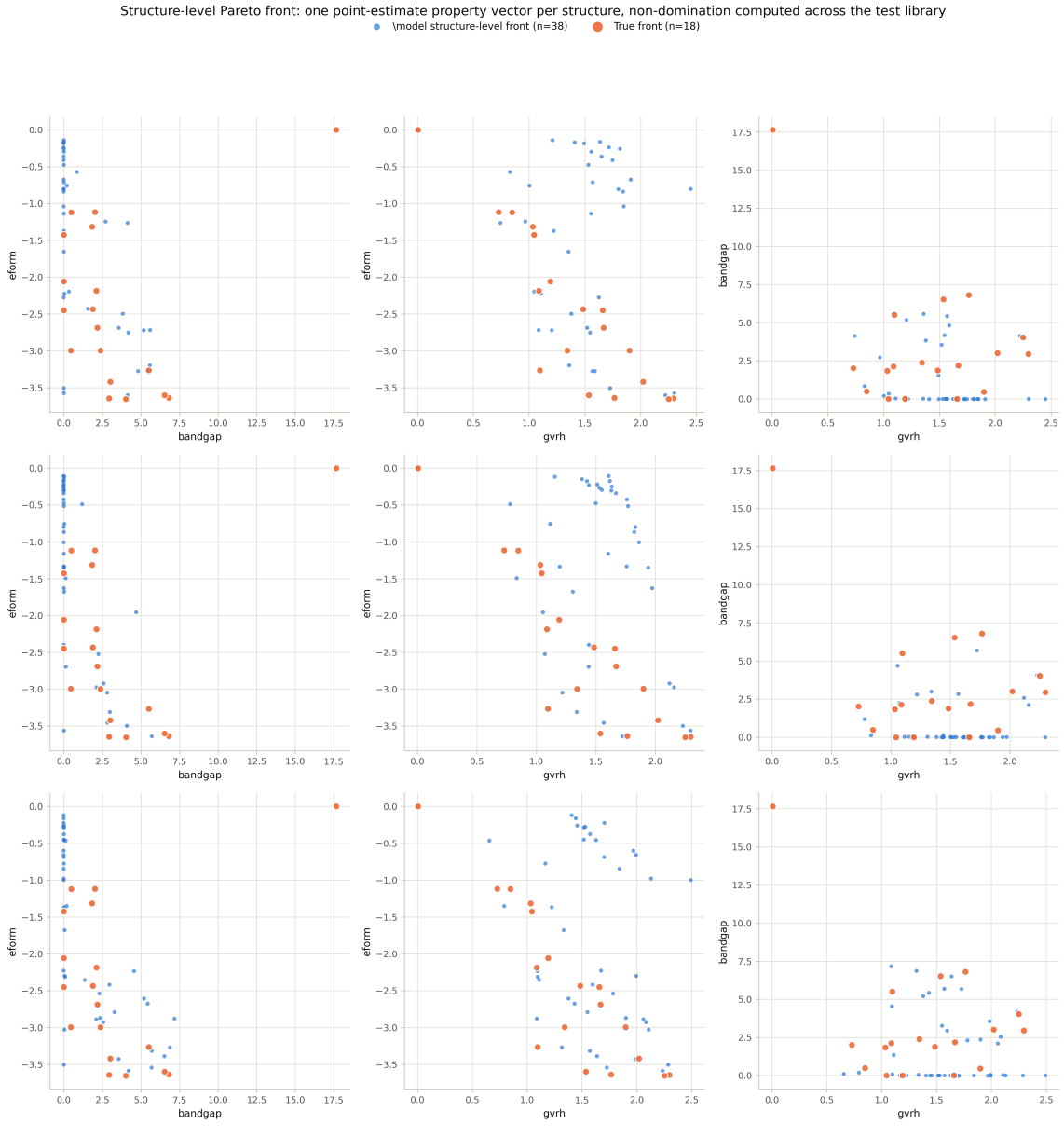


Fig. 3. Structure-level Pareto front: one point-estimate property vector per test structure (blue, n=38/39/42 for seeds 0/1/2) against the true non-dominated front computed from the actual DFT-measured properties of the same 250 test structures (orange, n=18, identical in every row by construction), shown as all three pairwise projections of the joint (eform, band gap, K_{VRH}) space. The single orange point at $E_g \approx 17.7$ eV is a genuine outlier near the dataset's documented maximum band gap (17.65 eV, Section 5); no model-front structure comes close to it in any of the three seeds. The model's front is visibly concentrated near zero band gap in every row, in contrast to the true front's wider spread.

some test structures' *true* band gap is high. The two points separate cleanly by cause: the DFO bounds of Eq. (10) are $[-2.470, 8.164]$ in normalised units, i.e. roughly $[-2.8, +12.2]$ eV once denormalised, so the ≈17.7 eV outlier is genuinely outside the region the DFO ever searches (the same search-bound limitation identified for K_{VRH} in the earlier, per-property analysis), while the ≈6.8 eV point sits comfortably *inside* the searchable range — yet no structure's point estimate reaches it. For that point specifically, the limitation is squarely the point estimate the trained heads return, not where the DFO is allowed to look.

In the eform–K_{VRH} and K_{VRH}–bandgap panels, the test set's K_{VRH} values span only ≈0–2.5 in $\log_{10}$GPa (≈1–300 GPa) — none of the extreme elasticity-data artefacts discussed in Section 2 (up to 3.8×10^{10} GPa) fall in this particular 250-structure test split; that outlier is in the training set instead (Section 2), so it does not affect this comparison. Over this physically reasonable K_{VRH} range, the model's front tracks the true front's shape reasonably well.

Practically, MEX-Multi's Pareto-front proposals are a reasonable guide to the well-populated, near-metallic part of the trade-off surface, but should not be read as covering the wide-band-gap end of it. This is the same caveat the formation-energy-versus-bulk-modulus asymmetry in Section 6.5 raises from a different angle: MEX-Multi's coverage of property space is uneven rather than uniform, and both the retrieval and the Pareto-design results should be trusted more where the model has demonstrated it, not assumed uniformly across the whole property range.

6.5 Embedding-Space Retrieval and Inverse Design Validation

The Pareto results test what MEX-Multi proposes; they do not test whether the shared embedding that produced those proposals carries property-relevant structure that generalises beyond MP20 at all. We test that directly here.

A key claim of MEX-Multi is that the shared equivariant backbone ϕ learns a property-aware embedding: geometrically similar structures in ϕ-space should also be similar in property space. We validate this claim by performing embedding-space retrieval against the Alexandria Materials Project (alex_mp20) database [12], a large-scale DFT-curated collection of inorganic crystals that is disjoint from the MP20 training, validation, and test splits.

6.5.1 Retrieval procedure. For each of the 250 test structures, we compute its embedding $\mathbf{x}_i = \phi(\mathcal{G}_i)$ using the trained backbone. We also embed every structure in alex_mp20 that has at least one DFT-computed property (formation energy, band gap, or bulk modulus), yielding approximately 34,500 candidate embeddings after removing any structure whose `material_id` overlaps with the MP20 train/val/test splits. For each test structure we retrieve its top-K=10 nearest neighbours in embedding space by cosine similarity. [**TODO:** strengthen the disjointness argument: Alexandria uses its own identifiers (`agm-*`), so `material_id` filtering will not remove Alexandria duplicates of MP structures. Dedupe by structure match (pymatgen `StructureMatcher`) or at minimum by reduced composition + space group, and report how many candidates were removed]

We compare five retrieval methods: a *random* baseline (uniform sampling from the alex_mp20 pool), three *single-property* MEX models (each trained on only one of formation energy, band gap, or K_{VRH}, so their embedding spaces are never exposed to the other two properties), and MEX-Multi. Comparing against the single-property embeddings, not only against random, isolates whether joint multi-property training is actually necessary for good retrieval, or whether any property-supervised embedding would do. Each method is trained/sampled with 3 independent seeds; rather than reporting one seed at a time, we pool the top-10 matches from all 3 seeds into a single candidate set per method before scoring, which reduces the variance a single noisy draw would otherwise introduce (this matters most for the random baseline, whose single-seed KL estimates swing considerably — see the Interpretation below).

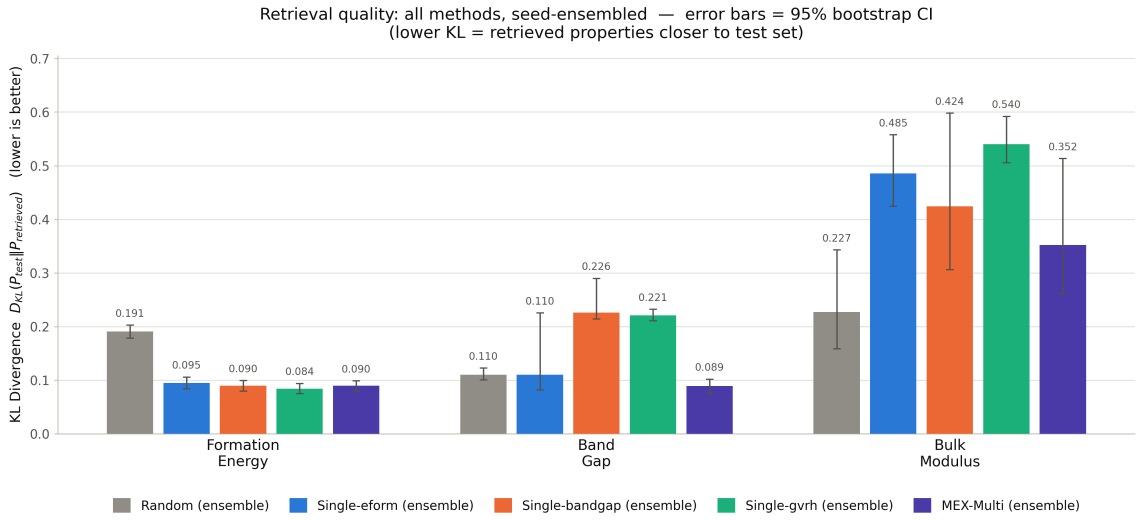


Fig. 4. Retrieval quality across all methods, seed-ensembled. For each method, top-k matches from all 3 training/sampling seeds are pooled into a single candidate set before computing the KL divergence $D_{\mathrm{KL}}(P_{\mathrm{test}} \,\|\, P_{\mathrm{retrieved}})$ between the retrieved and test-set property distributions (lower is better). Error bars denote 95% confidence intervals from 200 bootstrap resamples of the pooled candidate set. MEX-Multi retrieves distributions markedly closer to the test set than random retrieval and the single-property baselines for formation energy and band gap; for bulk modulus, random retrieval is competitive with (and here slightly better than) MEX-Multi once both are ensembled on equal footing, suggesting the learned embedding space is less informative for this property than for the other two.

6.5.2 Distributional alignment via KL divergence. Rather than inspecting raw histograms, we summarise retrieval quality as the KL divergence $D_{\mathrm{KL}}(P_{\mathrm{test}} \| P_{\mathrm{retrieved}})$ between the test-set property distribution and the retrieved pool's property distribution, for each of the five methods and all three properties. Lower values indicate the retrieved candidates are distributionally closer to the held-out test set. Figure 4 reports these values with 95% confidence intervals from 200 bootstrap resamples of each pooled candidate set, making explicit whether an apparent win over random retrieval is real or within noise.

6.5.3 Interpretation. The three panels of Figure 4 tell a property-dependent story rather than a uniformly positive one:

- **Formation energy:** MEX-Multi achieves the lowest KL divergence (0.090) alongside the single-bandgap (0.090) and single-gvrh (0.084) embeddings, all clearly below random (0.191). Any of the property-supervised embeddings — not only the one trained directly on formation energy (0.095) — retrieves formation-energy-aligned candidates well, suggesting stability-relevant structural information is shared broadly across these learned embedding spaces.
- **Band gap:** MEX-Multi is the clear best (0.089). Notably, the single-bandgap (0.226) and single-gvrh (0.221) embeddings are *worse* than random (0.110) at retrieving band-gap-aligned candidates, and single-eform (0.110) merely ties random. Training an embedding on one property does not by itself make it a useful proxy for band gap similarity — only the jointly-trained MEX-Multi embedding clears the random baseline here, which is direct evidence that joint multi-property training, not just property supervision in general, is what helps.

- **Bulk modulus:** this property inverts the pattern above. All four learned embeddings (single-eform 0.485, single-bandgap 0.424, single-gvrh 0.540, MEX-Multi 0.352) are *worse* than random retrieval (0.227) once random is pooled across seeds on equal footing. MEX-Multi is the best of the learned embeddings but does not beat chance, indicating the shared backbone has not learned structural features that are informative for K_{VRH} to the same extent as for formation energy or band gap.

These results show that MEX-Multi's embedding space transfers stability- and band-gap-relevant structural information to an entirely held-out database, and that for band gap specifically this transfer requires joint multi-property training rather than any single-property supervision. The bulk modulus result is a genuine limitation rather than a data artefact: despite MEX-Multi attaining strong *predictive* accuracy for K_{VRH} (Table 2), its embedding geometry is not yet a reliable retrieval signal for that property, so a practitioner using embedding-space retrieval for inverse design should trust MEX-Multi's rankings for formation energy and band gap targets more than for bulk modulus targets, pending further work on the retrieval geometry for mechanical properties.

7 Conclusion

We have presented MEX-Multi, an extension of the Matching-based EXtrapolation framework to the multi-property setting. Rather than regressing each target in isolation, MEX-Multi equips every property with an independent energy-based head over a single shared SE(3)-equivariant backbone, so that a single representation is shaped simultaneously by all K contrastive objectives while each target keeps its own label encoder, scoring network and negative-sampling regime. On three simultaneous targets drawn from the Materials Project MP20 collection, the model attains mean absolute errors of 0.110 eV/atom, 0.381 eV and 0.087 $\log_{10}$GPa, with Spearman correlations of 0.957, 0.716 and 0.969 (Table 2).

The clearest evidence for the central design choice is the controlled comparison in Table 3: holding backbone, energy-head architecture, loss functions, training procedure, data split and random seeds fixed, and varying only whether the model is trained on one property or three, joint training improves every metric for every property, reducing MAE by 68%, 42% and 59% for formation energy, band gap and bulk modulus respectively. Because capacity and training budget are held constant, the gain is attributable to the correlations the shared representation is able to exploit, which is precisely the regime—small per-property sample sizes over correlated targets—that motivates extending MEX beyond a single objective. A secondary finding is that the negative sampling distribution must be matched to each property individually: locally sampled negatives shape a sharp energy peak but leave the landscape undefined far from the ground truth, and for a target whose normalised range spans roughly 23 units this is enough to make the derivative-free optimiser fail at inference. Uniformly drawn InfoNCE negatives resolve this without altering the treatment of the narrower-range properties.

Beyond point prediction, we showed that the same optimiser, run repeatedly and followed by non-dominated sorting, exposes Pareto-optimal trade-off surfaces that can be compared directly against the true DFT-measured front, and that the learned embedding transfers to an entirely held-out database: retrieval against alex_mp20 recovers property distributions closer to the test set than random retrieval for formation energy and band gap, and does so where single-property supervision alone does not.

Our results also delimit where the method should not yet be trusted. Coverage of property space is uneven: candidates thin out toward the most stable end of the formation-energy range and, in all three seeds, fail entirely to reach a held-out material near the dataset's maximum band gap—a direct consequence of anchoring the search interval to

training-set extremes rather than to test-time outliers (Eq. (10)). More pointedly, retrieval for bulk modulus does not beat a random baseline once both are ensembled on equal footing, despite strong predictive accuracy for that same property. Predictive accuracy and embedding geometry are therefore not interchangeable evidence, and a practitioner using MEX-Multi for retrieval-driven design should weight its formation-energy and band-gap rankings more heavily than its mechanical ones.

Three directions follow naturally. First, a single-task baseline retrained on the identical 5k split would place the absolute numbers in Table 2 against published architectures on equal footing, a comparison we deliberately declined to make against figures obtained on the full MP20 split. Second, search bounds that adapt per structure, rather than a fixed buffer around global training extremes, would address the outlier-coverage gap without abandoning the stability that motivated the current choice. Third, and most substantively, the bulk-modulus retrieval result suggests that the structural features governing mechanical response are underrepresented in the shared embedding; whether this reflects the property's weaker dependence on local coordination environments, the smaller number of labelled examples, or a limitation of shared-trunk coupling itself is an open question, and answering it would strengthen the case for energy-based matching as a general-purpose route to multi-property inverse design. A fourth direction is suggested by what MEX-Multi deliberately does not do: because the shared embedding is never conditioned on observed property values, inter-property information reaches it only as a gradient. Feeding the encodings of the other $K - 1$ targets into the representation, with the label dropout needed to keep training and inference inputs consistent, is a natural next step, and the comparison against the present architecture would isolate how much is gained by moving that coupling from the loss into the input.

[**TODO:** add a sentence on the InfoNCE ablation (Section 6.3) once it is run]

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